Supply Chain Demand Forecasting

Multi-Product Prediction & Inventory Optimization

Felipe Taha Sant'Ana

March 2026

Agenda

1. Business Context
2. EDA
3. Model Results
4. Inventory Optimization
5. Conclusions

The Supply Chain Challenge

Problem

- Over-stocking ties up **15–30%** of working capital
- Stockouts lose **4–8%** of annual revenue
- Manual planning cannot scale across products and warehouses

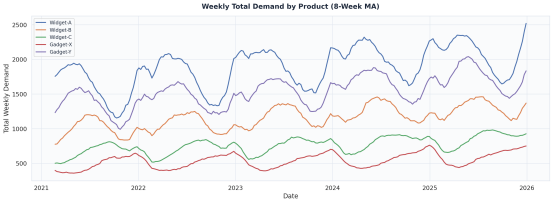
Dataset

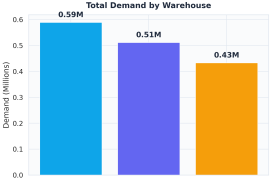
Records	3,915
Products	5
Warehouses	3
Period	5 years
Frequency	Weekly

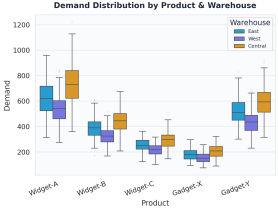
Our Approach

ML-driven demand forecasting across **5 products** $\times$ **3 warehouses**, with safety stock optimization for cost-service trade-off.

Demand Trends & Warehouse Distribution



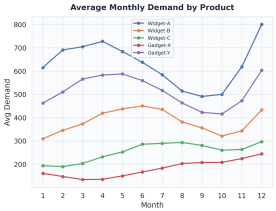


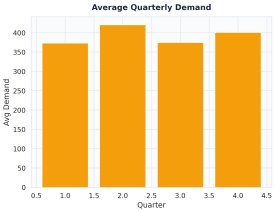


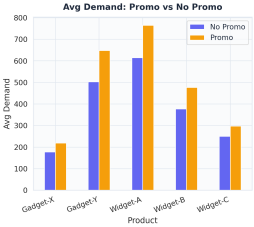
Findings

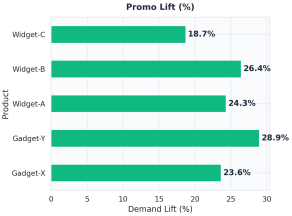
Clear upward trends with strong seasonal patterns. Central warehouse handles the highest volume, followed by East and West.

Seasonality & Promotional Effects





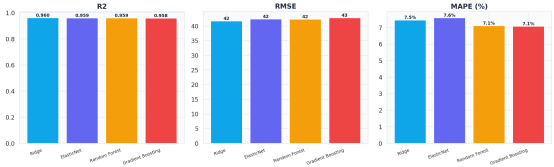


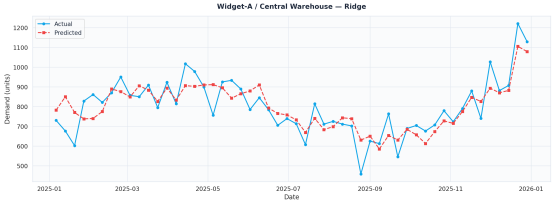


Key Insights

Each product has a unique seasonal peak. Promotions drive a **25–35% demand lift** across all products — critical for inventory planning.

Model Comparison & Forecast

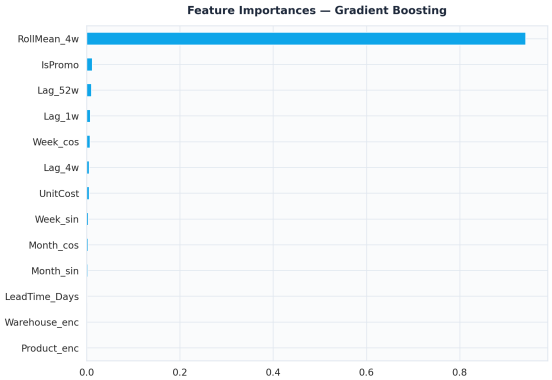


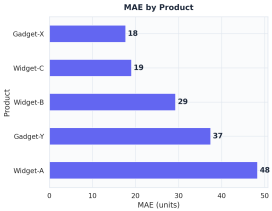


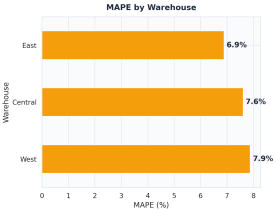
All Models Achieve $R^2 > 0.95$

Model	R^2	RMSE	MAPE
Random Forest	0.959	42.3	7.1%
Gradient Boosting	0.958	42.7	7.1%
Ridge	0.957	43.4	7.5%

Feature Importance & Error Analysis



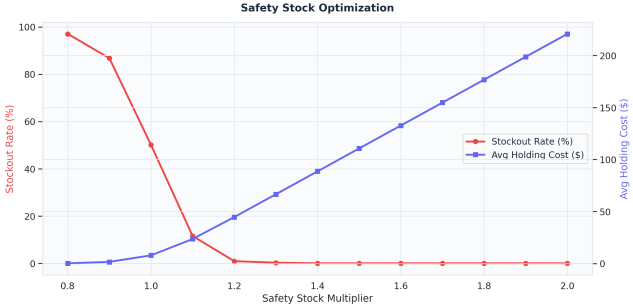




Key Drivers

Lag features (1-week, 4-week) are dominant predictors. Product identity and seasonal encodings provide additional signal. Higher-volume products naturally have higher absolute error.

Safety Stock Optimization



Trade-off Analysis

- 1.0x High stockout risk (15%+)
- 1.2x Balanced (5% stockout, moderate cost)
- 1.5x Conservative (2% stockout, higher cost)
- 2.0x Over-stocked, minimal stockouts

Recommendation

1.2–1.4x safety stock multiplier for optimal cost-service balance.

Summary & Future Work

Achievements

- $R^2 > 0.95$, MAPE $\sim$7%
- Multi-product, multi-warehouse model
- Safety stock optimization framework
- Promo-aware forecasting

Future Work

1. Prophet / NeuralProphet
2. Probabilistic forecasts (quantile)
3. Multi-echelon optimization
4. Demand sensing with POS data
5. Automated reorder triggers

Thank You

Questions & Discussion

GitHub: github.com/ftahas
Email: ftahas@proton.me
LinkedIn: linkedin.com/in/ftsantana